

IT Research Methodology

Quantitative Data Analysis and Interpretation

Image Source: Research Methodology

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Outline

- Steps in the Process of Quantitative Data Analysis
- Data Preparation for Analysis
- Data Analysis
- How Do You Report the Result?
- How Do You Interpret the Results?

❖ Steps in the Process of Quantitative Data Analysis

➢ There are several interrelated steps used in the process of analyzing quantitative data.

1. **Data Preparation for Analysis**: This involves determining how to assign numeric scores to the data, assessing the types of scores to use, selecting a statistical program, inputting the data into a program, and then cleaning up the database for analysis

❖ Steps in the Process of Quantitative Data Analysis

➢ There are several interrelated steps used in the process of analyzing quantitative data (Cont.).

2. **Data Analysis**: Typically, you conduct a descriptive analysis of the data-reporting measures of central tendency and variation. Then you conduct more sophisticated inferential analysis to test hypotheses and examine confidence intervals and effect sizes.

❖ Steps in the Process of Quantitative Data Analysis

➢ There are several interrelated steps used in the process of analyzing quantitative data (Cont.).

3. **Result Report**: the results that are found using tables, figures, and a discussion of the key results
4. **Result Interpretation**: you interpret the results from the data analysis. This consists of summarizing the results, comparing the results with past literature and theories, advancing the limitations of the study, and ending with suggestions for future research.

❖ Data Preparation for Analysis

➢ **Preparing and organizing data for analysis** in quantitative research consists of

- ✓ Scoring the data and creating a codebook
- ✓ Determining the types of scores to use
- ✓ Selecting a computer program
- ✓ Inputting the data into the program for analysis, and
- ✓ Cleaning the data.

❖ Data Preparation for Analysis

➢ **Scoring Data** means that the researcher assigns a numeric score (or value) to each response category for each question on the instruments used to collect data.

Please check the appropriate response to this statement:
"Students should be given an opportunity to select a school of their choice."

_____ Strongly agree
_____ Agree
_____ Undecided
_____ Disagree
_____ Strongly disagree

❖ Data Preparation for Analysis

➢ To make scoring easy, you can pre-assign numbers on the instrument to each response option, such as in this example:

Please respond to this question:

"Fourth graders should be tested for math proficiency."

_____ (5) Strongly agree
_____ (4) Agree
_____ (3) Undecided
_____ (2) Disagree
_____ (1) Strongly disagree

❖ Data Preparation for Analysis

➢ Scoring Data - Codebook

Codebook for High School Smoking Project	
Variable 1.	ID—identification number assigned to each student; from 1–50
Variable 2.	Gender—sex of the student; 1 = Male, 2 = Female
Variable 3.	Grade—grade level of the student; 10 = 10th grade, 11 = 11th grade, 12 = 12th grade
Variable 4.	Parents—parents' marital status; 1 = married, 2 = divorced, 3 = separated
Variable 5.	Smoke—whether the student smokes or not; 1 = no; 2 = yes
Variable 6.	Chewer—whether student chews tobacco; 1 = no; 2 = yes
Variable 7.	Ability—academic ability based on grade point average last semester; 1 = below 2.0; 2 = 2.0–2.9; 3 = 3.0–3.5; 4 = 3.6–4.0
Variable 8.	Depression—total depression score for all items on an instrument measuring depression; scores from 20 to 100

❖ Data Preparation for Analysis

➢ Determine the Types of Scores to Analyze:

Types of Scores Used in Quantitative Analysis			
	Single-Item Scores*		
	Question 1 Score	Question 2 Score	Question 3 Score
Jane	5	4	3
Jim	4	3	4
John	2	1	2
Jean	4	5	4
Julie	4	3	4
Johanna	5	4	5

❖ Data Preparation for Analysis

➢ Determine the Types of Scores to Analyze (Cont.):

	Summed Score or Scale*					
	Question 1	Question 2	Question 3	Question 4	Question 5	Summed Scores
Jane	5	4	3	4	4	20
Jim	4	3	4	4	3	18
John	2	1	2	2	3	10

	Net or Difference Scores		
	Pretest Math Score	Posttest Math Score	Net or Difference Score
Jane	80	85	+5
Jim	76	77	+1
John	65	75	+10
Jean	95	97	+2
Julie	91	94	+3
Johanna	93	95	+2

*Question response scale is: 5 = strongly agree, 4 = agree, 3 = undecided, 2 = disagree, and 1 = strongly disagree.

❖ Data Analysis

➢ Questions or hypotheses in quantitative research require that you do the following:

- ❖ Describe trends in the data for a single variable or question on your instrument (e.g., “What is the self-esteem of middle school students?”).
  - ✓ **Descriptive statistics** that indicate general tendencies in the data (**mean, median, and mode**); the spread of scores (**variance, standard deviation, and range**); or a comparison of how one score relates to all others (**z scores or percentile rank**).

❖ Data Analysis

➢ Questions or hypotheses in quantitative research require that you do the following (Cont.):

- ❖ Compare two or more groups on the independent variable in terms of the dependent variable (e.g., “How do boys and girls compare in their self-esteem?”).
  - ✓ **Inferential statistics** in which we analyze data from a sample to draw conclusions about an unknown population

❖ Data Analysis

➢ Questions or hypotheses in quantitative research require that you do the following (Cont.):

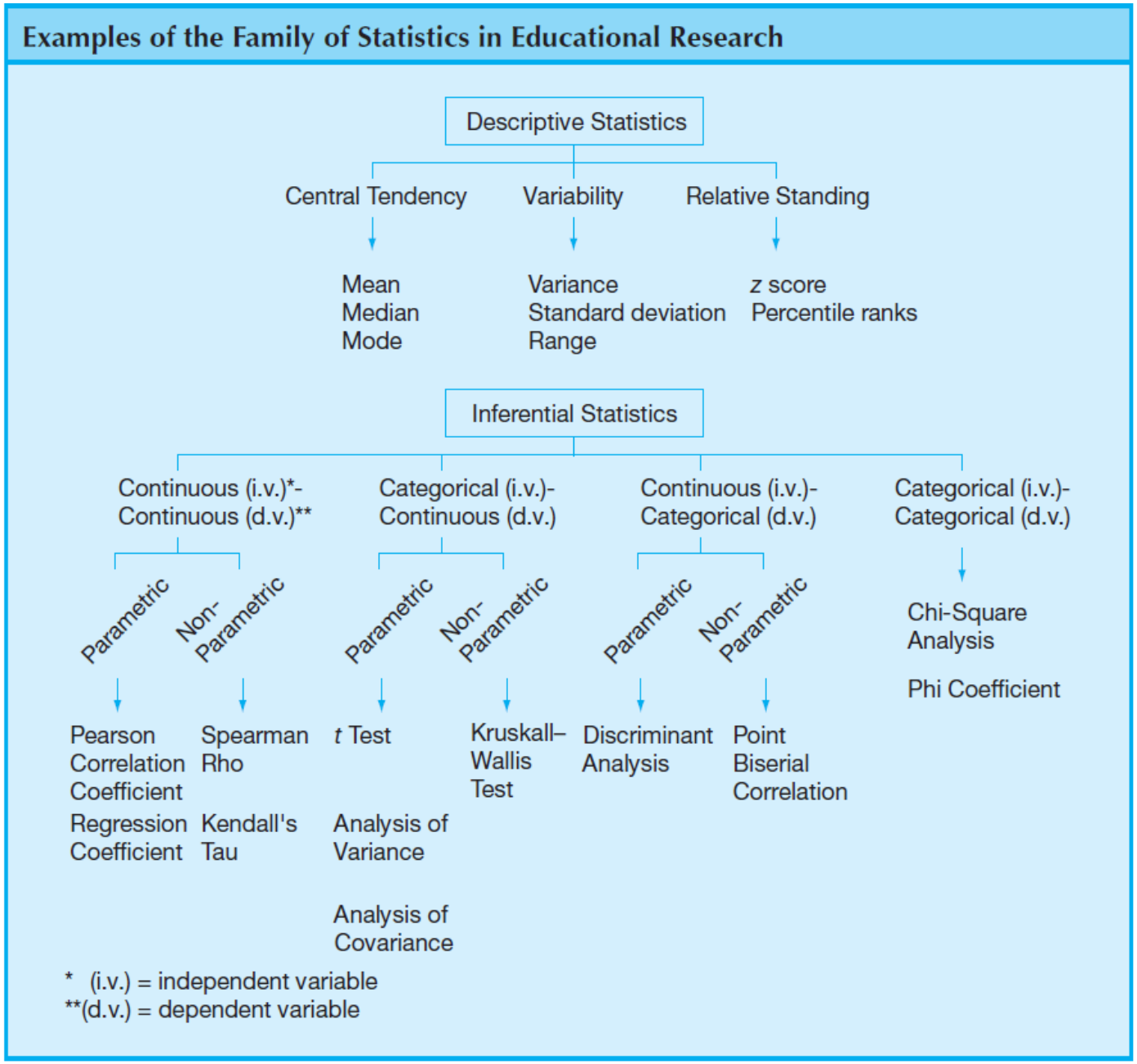
- ❖ Relate two or more variables (e.g., “Does self-esteem relate to an optimistic attitude?”), we also use inferential statistics.

❖ Data Analysis

➢ Questions or hypotheses in quantitative research require that you do the following (Cont.):

- ❖ Examine the **relationship among variables** through more advanced statistical procedures, such as *regression analysis*, *meta-analysis*, *factors analysis*, *discriminant function analysis*, *path analysis*, *structural equation modeling* or *hierarchical linear modeling*.

Data Analysis



❖ Data Analysis

➢ Example of the three measures of central tendency for 10 students for whom we have depression scores

Descriptive Statistics for Depression Scores

Raw Scores	*z* Score	Rank
60	−1.57	10.0
64	−1.28	20.0
75	− .49	30.0
76	− .42	50.0
76	− .42	50.0
83	+ .09	60.0
93	+ .81	70.0
94	+ .92	80.0
98	+1.22	90.0
99	+1.24	100.0

Sum = 818

Mode = 76

Median = 79.5

Mean (M) = 81.8

Variance (standard deviation [SD^2]) = $\sum \frac{(\text{raw score} - M)^2}{N - 1}$

Variance = 193.29

$SD = \sqrt{\text{variance}}$

SD = 13.90

z score = raw score − M/SD

Range: minimum = 60; maximum = 99

❖ Data Analysis

➢ Measures of variability indicate the spread of the scores in a distribution.

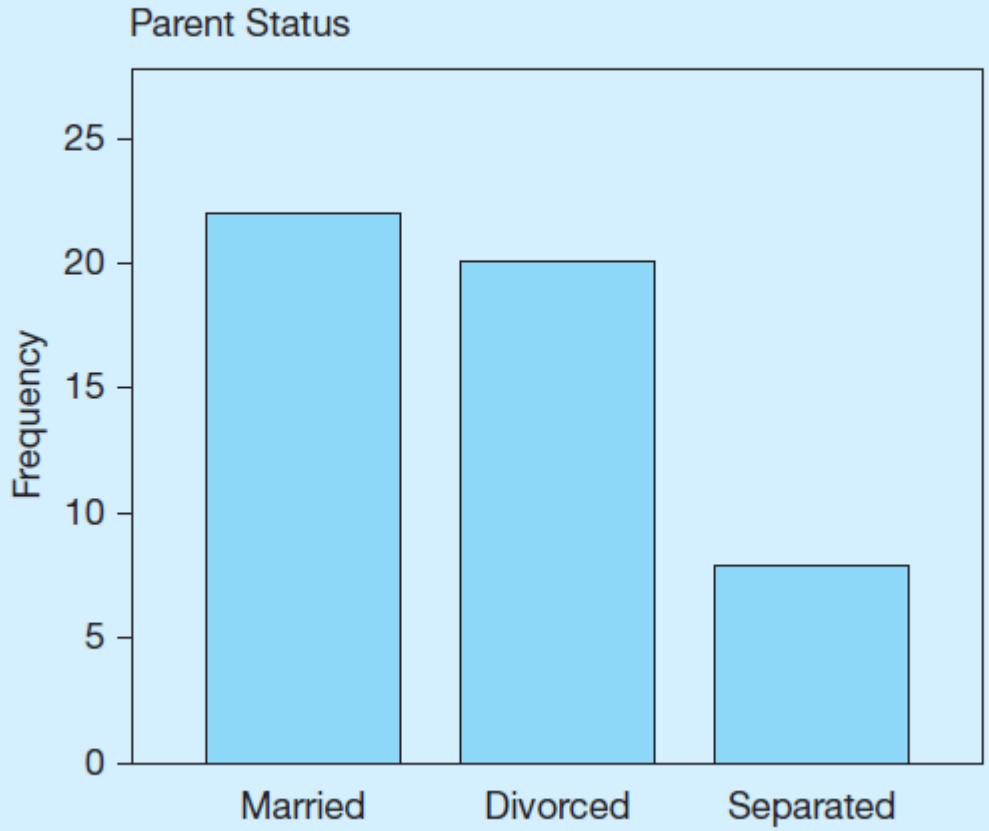
Range, variance, and standard deviation all indicate the amount of variability in a distribution of scores.

Descriptive Statistics for the Categorical Variable "Parent Status"

	Parent Status			
	Frequency	Percent	Valid Percent	Cumulative Percent
Married	22	44.0	44.0	44.0
Divorced	20	40.0	40.0	84.0
Separated	13	16.0	16.0	100.00
Total	50	100.0	100.0	

❖ Data Analysis

Bar Chart Depicting the Frequencies for the Categorical Variable "Parent Status"



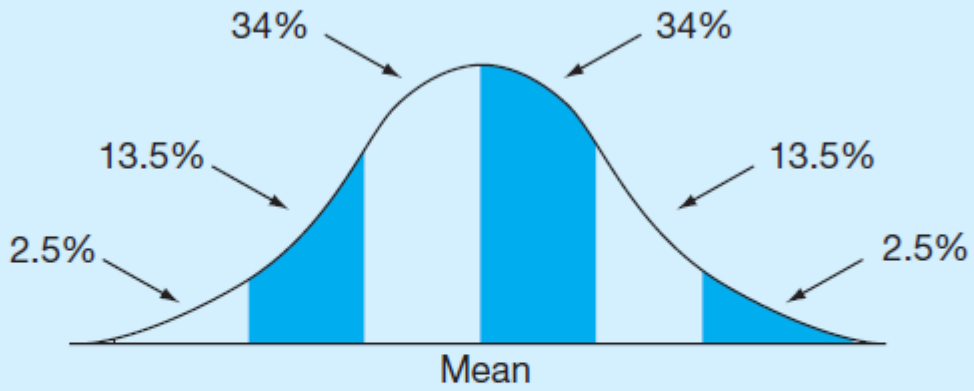
❖ Data Analysis

➢ The variance indicates the dispersion of scores around the mean. Calculating this score is easy:

- ❖ Find the difference between the mean and the raw score for each individual
- ❖ Square this value for each individual
- ❖ Sum these squared scores for all individuals
- ❖ Divide by the total number of individuals minus 1

Data Analysis

The Normal Curve



SD, *z* scores	-3	-2	-1	0	+1	+2	+3
Percentiles	0.13	2.28	15.87	50	84.13	97.72	99.87
t scores	20	30	40	50	60	70	80

Source: Adapted from Gravetter and Wallnau (2013).

❖ How Do You Report the Result?

➢ A results section includes the following:

✓ Tables that summarize statistical information

✓ Figures (charts, pictures, or drawings) that portray variables and their relationships

✓ Detailed explanations about the statistical results

❖ How Do You Interpret the Results?

➢ After reporting and explaining the detailed results, researchers conclude a study by

1. Summarizing key findings
2. Developing explanations for results
3. Suggesting limitations in the research, and
4. Making recommendations for future inquiries.

Q&A